

Supplementary: Web Application Development

Information Security

Application classifications

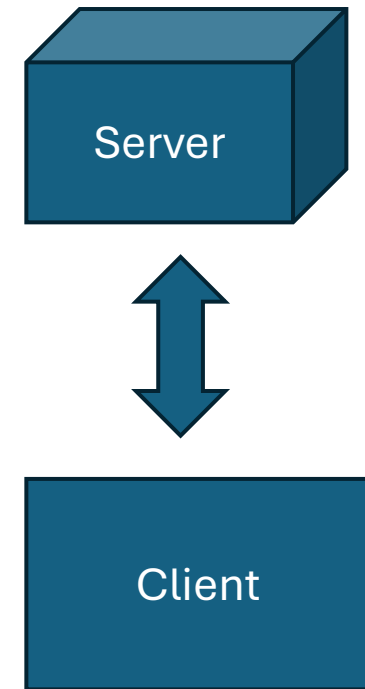
- Applications are different in many aspects:
 - Purposes
 - Users
 - *Runtime Environments*

Types of applications (by environment)

- Native Applications
 - Desktop Applications
 - Mobile Applications
- Scripting applications
- **Web Applications**
 - *Most important type of applications nowadays*

Web Application overview

- Client – Server
- Client : provide interactive user-facing interface (GUI)
- Server:
 - Provide logic according to user interactions
 - Store user data (centralized data management)



Why Web applications are IMPORTANT ?

- Ease of access and usage (for end users)
 - No software installation (just need browser)
 - Very nice GUI widgets
 - Very lightweight
- Ease of deployment and management (for service provider):
 - Centralized. Deploy once, serve many
 - Data centralization
- Excellent technology stacks
 - Highly customizable, flexible, reusable, easy to learn, quick in development

Web application technologies

- A web application = 2 programs + running at 2 separated locations
 - Frontend and backend
- Frontend: RUN INSIDE browser
- Backend: RUN INSIDE server
- Both are scripting technology

Frontend technologies

- **ONE** single standardized stack:
 - HTML: for representing content
 - CSS: for styling content
 - JavaScript: for handling client logics (animation, request sending, data visualization, user input)
- **AGAIN: Run within the BROWSER!**
 - Runtime environment = BROWSER
- **DOM API: set of functions to interact with BROWSER programmatically**

Backend Technologies

- MANY options to choose
 - Java, PHP, NodeJS, ... <-- called "*application server*"
 - RDBMS, noSQL <-- called "*database server*"
 - Supplementary components : redis , memcache, ...
- Easy to scale:
 - Easy to use load balancers : Haproxy, nginx/apache
- Optional only :
 - But are expected since it provides features.

Frontend – Backend Interaction

- Over HTTP
- Request – Reply cycles
- Each Request – Reply $\Leftrightarrow$ 1 URL
- Each Request – Reply $\Leftrightarrow$ 1 page / web object (as output)
- Example:
 - Client sends a Request (for an URL) --> backend produces output for the URL --> send the output in a Reply --> client views Reply content (render a page)

Backend Request Processing

- Backend processes request by request
 - Url by url
- No "main" function, there only multiple "request handlers"
 - Request handler = function that process a request
- Typical tasks:
 - Parsing request to get parameters (query string, headers, form, cookies, ...)
 - Query DB to fetch data or modify data in the DB
 - Format the queried data properly to make the reply

Frontend processing

- Actively sends requests
 - Manually type url in address bar
 - Javascript code sends requests (client logics): e.g. on click a button
- Wait for a reply (after sending a request)
- Process the reply:
 - Render a page
 - Manipulate the current page (AJAX)

Components of a Web Application Project

- A Web application may have:
 - Sub projects for backend + Sub projects for frontend
 - A single project that houses both frontend and backend
 - A single project for backend only (frontend code is automatically generated)
 - A single project for frontend only (static web, no backend)
- Projects' source code are managed by "source version control" (e.g.: Git)
 - GitHub/GitLab/Gerrit
- Team collaboration is supported and expected.

Frontend development skills

- Minimalist:
 - HTML + CSS
 - Javascript + DOM API
- Intermediate to Advance:
 - Minimalist and ...
 - React/Angular/...

Backend Development Skills

- Choose a stack and master it:
 - PHP
 - Java
 - Python
 - NodeJS
 - ...
- SQL and database management
- Others : noSQL, Cache, load balancing, ...

Key skill: Control your ENVIRONMENTS

- Mastering your browser:
 - Developer Tools:
 - Console/Network/Element/Style
- Mastering your application server
 - How to install your stack: apt-get
 - How to configure your application server
 - edit configuration files
 - How to run your application server
 - Run app, systemctl , ...
- Mastering your DB server, webserver and supplementary servers
- Mastering Git-based collaboration procedure

Lab challenge 1: Static Webserver

- Controlling nginx webserver
 - Question set 1:
 - What are command to install nginx?
 - After installation, what are commands to stop/start nginx?
 - After running nginx, how to show nginx's processes ?
 - After running nginx, how to check service ports that nginx is listening on?
 - Question set 2:
 - Where are configuration files for nginx ?
 - How many virtual servers are there by default?
 - Where is the default "document root"?
 - What is user account for running nginx ?
 - Create a new folder and make it the new "document root". Show commands and notices.

Lab challenge 2: Static Web on nginx

- Edit a simple HTML document and store it in "document root"
- Load the document via your browser.
- Questions:
 - Show content of your html
 - Show how to copy the file to nginx document root
 - What ownership and mode should you set to your html file ? Show command to change mode and ownership
 - Show screenshot that proves the html file is successfully loaded.
 - Change your html document content to add an image to the page. Show screenshot to prove your success.

Lab 3: Write Javascript code

- Go to this page and follows all the tutorial:
 - <https://www.w3schools.com/js/>

Lab 4: PHP-FPM application server

- Controlling PHP-FPM application server
 - Question set 1:
 - What are command to install php-fpm?
 - After installation, what are commands to stop/start php-fpm?
 - After running , how to show php-fpm's processes ?
 - After running, how to check service ports that php-fpm is listening on?
 - Question set 2:
 - What is user account for running php-fpm ?
 - User Google/GenAI to have nginx configuration for integrating with PHP-FPM. Show procedure of integration.
 - Test the stack (Nginx + PHP-FPM) with helloworld php file. Show the procedure.
 - Show PHP environment information with phpinfo() function

Control PHP environment

- Configure Nginx for PHP
- First PHP application
- PHP template (PHP tag):
 - Dynamic content vs static content
 - PHP execution and JavaScript execution
 - NEED to know what is run where
- PHP output with print()
- Configure for debugging : php.ini configuration
 - *phpinfo()* function

[Demo] Request Handling

- Request method
 - GET: get data from server
 - POST: upload data to server
 - Detecting request method with superglobal `$_SERVER`
- GET Request handling:
 - Working with "Query String"

[Demo] Request Handling: POST request

- Working with form
 - FORM action
 - Input element
- Process POST request

[Demo] Request Handling: Cookies

- How cookie work
 - Set-Cookie and Cookie headers
- header() function
- setcookie() function
- \$_COOKIE superglobal

[Demo] Request Handling: Session in PHP

- Understand session in PHP
 - `session_start()`
 - `session_id()`
 - `session_status()`
 - `session_destroy()`

Sending request from browser

- Send request from address bar
- Send request with HTML
 - using FORM
 - Using other elements: `<img>`, `<script>`
- Send request with JavaScript
 - `fetch()` function

MySQL installation

- Install mysql-server in Ubuntu
- Access mysql server with *mysql* tool
- Create an account and grant privileges
- Install adminer

Working with database

- Reconfigure mysql server for listening at a network socket
- Practice:
 - Create a new database
 - Create a new user
 - Grant privileges

Working with database

- Create/Design tables in a database
- "Account" table